Today:

- where was the size-biasing in age & residual life?

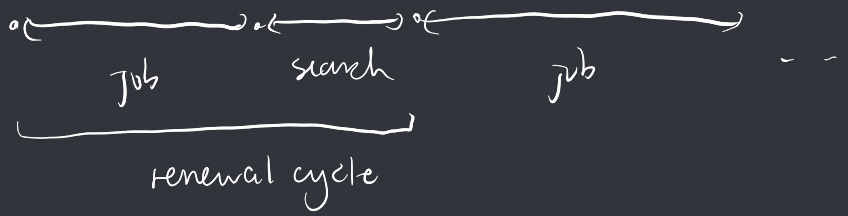
  (age + residual life is size-biased)

- exercises from ch. 3
- ch. 4: continuous-time Markov chains

Exercises, ch. 3

(2) mean job length = 11 mths.

search = exp w/ mean = 3 mths



$$\frac{\text{time spent in job}}{\text{total time}} = \frac{11}{11+3} = 11/14.$$

(4) arrivals with waiting time having distbn: F

w/ mean μ_F

random fares w/ mean μ_G.

W: total fares.

$$= \sum_{i=1}^{N(t)} r_i$$

$N(t)$: # arrivals to time t.

r_i: i^{th} fare

asymptotic $\dfrac{\text{total fares}}{\text{time}} = \underbrace{\dfrac{\text{fare}}{\text{taxi}}}_{=\mu_G} \cdot \dfrac{\text{taxi}}{\text{time}} = \dfrac{1}{\underbrace{\dfrac{\text{time}}{\text{taxi}}}_{=\mu_F}}$

$= \mu_G / \mu_F$

(6) $i \in \{1,2,3\}$, p_i to succeed.

Take turns shooting ball until failure, then switch

✓	✓	×	✓	✓	✓	×	×	✓	×	
1	1	1	2	2	2	2	3	1	1	...

$\underbrace{\quad}_{L_1}$ $\underbrace{\quad}_{L_2}$ $\underbrace{\quad}_{L_3}$ ↑ stop renewal cycle here. → n = # turns

(a) proportion of time each $i \in \{1,2,3\}$ spends playing.

Picture renewal cycle that stops after we're back to player 1.

Let L_i = time spent player i shuts in one cycle.

then

$$L_1, L_2, L_3$$

are independent & $L_i \overset{d}{=} \text{geo}(1-p_i)$

$$E[L_i] = \frac{1}{1-p_i}$$

Let $q_i = 1-p_i$ (failure prob.)

Then

$$\text{prop time player } i \text{ shuts} = \frac{E[L_i]}{\sum_{j=1}^{3} E[L_j]} = \frac{1/q_i}{\frac{1}{q_1}+\frac{1}{q_2}+\frac{1}{q_3}}$$

(b) if $p_1 = 2/3$ $p_2 = 3/4$ $p_3 = 4/5$:

$q_1 = 1/3$ $q_2 = 1/4$ $q_3 = 1/5$

player i shoots basketball for

i	fraction time
1	$3/12$
2	$4/12$
3	$5/12$

(9)

E time $= 1/\lambda$ μ_G

renewal cycle: $\underbrace{\text{arrival}}$, $\underbrace{\text{dealing}}$, ...

(a) rate of sales (#sales/time) $= \dfrac{1}{E[\text{length of cycle}]} = \dfrac{1}{\frac{1}{\lambda} + \mu_G}$

(b) fraction customers lost.

$E[\#\text{arrivals during absence}] = \lambda \mu_G$

waiting — dealing

#lost/cycle

#actual customers / cycle = 1

fraction lost $= \dfrac{\lambda \mu_G}{1 + \lambda \mu_G}$

age & residual life example

Is age & residual life for Poisson process (w/ intensity λ) distributed as $\exp(\lambda)$?

Found that both have marginal density function

$$g(z) = \frac{P(t_i > z)}{E\, t_i}$$

For exponential $P(t_i > z) = e^{-\lambda z}$ $\quad E\, t_i = 1/\lambda$

giving density

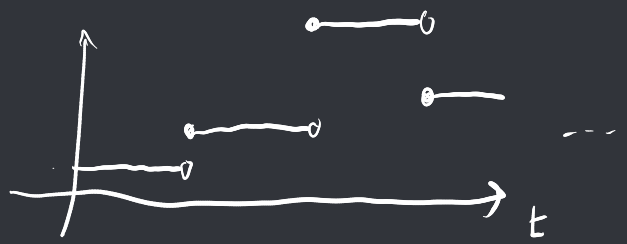
$$\lambda e^{-\lambda z}.$$

(converse)

challenge exercise: suppose $g(z)$ is the density function of t_i.

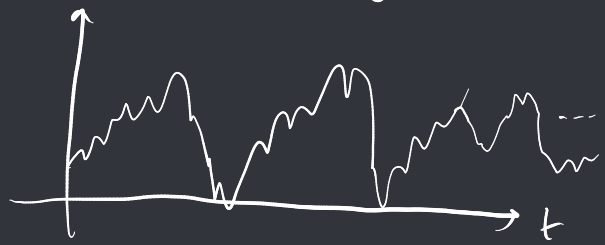
Show that t_i must have exponential distbn.

☆ Ch. 4 continuous-time Markov chains



continuous-time trajectory that is piecewise constant

We will study these



& not this (for now)

A Markov process in continuous time is $(X_t)_{t \geq 0}$ such that $\forall\ t_1 < \dots < t_n < t_{n+1}$, $\forall x, y$,

$$P(X_{t_{n+1}} = y \mid X_{t_n} = x, X_{t_{n-1}}, \dots, X_{t_1})$$

$$= P(X_{t_{n+1}} = y \mid X_{t_n} = x)$$

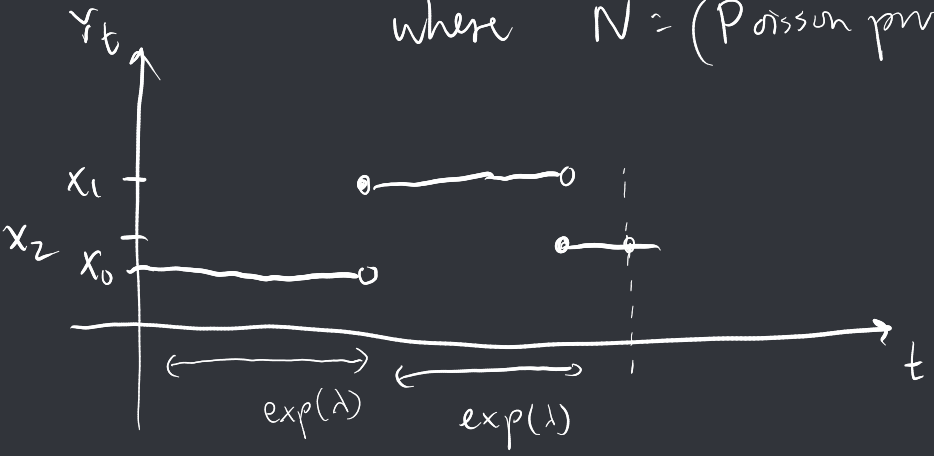
It is time-homogeneous if this

$$= P(X_{t_{n+1} - t_n} = y \mid X_0 = x)$$

<u>Example</u>: let (X_n) be a (discrete-time) Markov chain

and let $Y_t = X_{N(t)}$

where $N = (\text{Poisson process})(\lambda)$.



why does (Y_t) have Markov property?

knowing the value of Y_t for fixed t completely determines distbn. of $(Y_{t+s})_{s \in \mathbb{R}_+}$

(uses memoryless property of exponential and fact that (X_n) is a Markov chain).

Continuous-time piecewise constant Markov process $=$ continuous-time Markov chain (CTMC)

CTMCs are characterized by "transition rates"

$$\boxed{q(i,j) := \lim_{h \to 0} \frac{p_h(i,j)}{h} \qquad j \neq i}$$

where $p_t(i,j) = P(X_t = j \mid X_0 = i)$.

Ex: if $(X_t)_{t \in \mathbb{R}_+} \overset{(d)}{=} PP(\lambda)$ then

$$p_t(i,j) = P(\text{Poiss}(\lambda t) = j - i)$$
$$= e^{-\lambda t} (\lambda t)^{j-i} / (j-i)!$$

$$q(i,j) = \frac{d}{dt} p_t(i,j) \Big|_{t=0}$$

$$= \begin{cases} \boxed{0 \quad \text{if} \quad j \neq i+1} \\ \boxed{\lambda \quad \text{if} \quad j = i+1} \end{cases}$$

$$\frac{d}{dt}(\cdot) = \underbrace{\qquad\qquad}$$

$$-\lambda p_t(i,j) + p_t(i,j) \frac{(j-i)}{t}$$

(if $i \neq j$ then $p_0(i,j) = 0$)

$$P(\text{Poiss}(\lambda h) = 1) = \lambda h + O(h^2)$$

$$P(\text{Poiss}(\lambda h) \geq 2) = O(h^2)$$

$\therefore$ Poisson process has $q(i, i+1) = \lambda$

and $q(i,j) = 0$ other $j \neq i$.

Diagram:

$$\begin{tikzcd} \underset{0}{\circ} \arrow[r, "\lambda"] & \underset{1}{\circ} \arrow[r, "\lambda"] & \underset{2}{\circ} \arrow[r, "\lambda"] & \underset{3}{\circ} & \cdots \end{tikzcd}$$

Other example $Y_t = X_{N(t)}$;

use Poisson distbtn:

$$P\left(N(t+h) = N(t)+1 \right) = \lambda h + O(h^2)$$

for $y \neq x$

$$P\left(Y_{t+h} = y \mid Y_t = x \right) = P\left(\begin{matrix} N(t+h) = N(t)+1 \\ X_{N(t)+1} = y \end{matrix} \middle| Y_t = x \right)$$

$$+ \; O(h^2)$$

$$= P\left(X_{N(t)+1} = y \,\middle|\, N(t+h) = N(t)+1,\ Y_t = x\right)$$

$$P\left(N(t+h) = N(t)+1 \,\middle|\, Y_t = x\right) + O(h^2)$$

$$= p(x,y)\lambda h + O(h^2)$$

$$\therefore \quad q(x,y) = \lambda p(x,y)$$

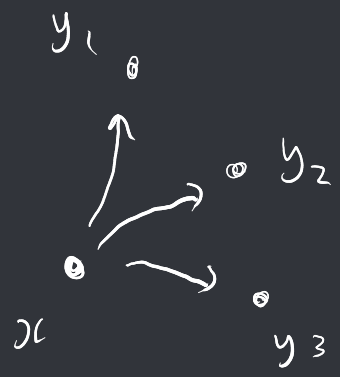
<u>Constructing</u> CTMC from set of transition rates.

Given S & $\underbrace{q(x,y)}_{\geq 0}$, $x, y \in S$ (some should be > 0)

how to obtain (Y_t) w/ Markov property & satisfies

$$(*) \quad \lim_{h \to 0} \frac{P(Y_{t+h} = y \mid Y_t = x)}{h} = q(x,y)?$$

"Constructing" = obtaining stochastic process (defined in terms of specific random variables) with required properties.



at each transition, attach exponential RVs w/ rates $q(x,y)$ for each y, that are independent:

$$(W_y)_{y \in S}$$

Let $W = \min_y W_y$ & $Y(t+W) = z$.

$$z = \operatorname{argmin}_y W_y$$

wait for first possible transition & take it.

Then (W, Z) are indep dist.

and

$$W \overset{d}{=} \exp\Big(\overbrace{\sum_{y \neq x} q(x,y)}^{\lambda(x)}\Big),$$

$$P(Z = y) = \frac{q(x,y)}{\lambda(x)}.$$

To construct the process, do this at each transition: Let $T_0 = 0$

$$T_{n+1} = T_n + \left(\text{waiting time for } n+1^{st} \text{ trans.}\right)$$

$$X_n = \text{value at } n^{th} \text{ transition}$$

then

$$Y_t = X_n \quad \forall \; T_n \le t < T_{n+1}$$

exercise: check that (*) holds at first transition (for $t=0$)

Transition semigroup $p_t(x,y)$, $x,y \in S$, $t \geq 0$

satisfies

- Chapman-Kolmogorov equations:

$$p_{t+s} = p_t p_s \quad \forall\, t,s$$

(as a matrix; Markov property)

- Kolmogorov forward and backward equations

$$p_t = e^{Qt}$$

(system of linear ODEs satisfied by (p_t))

$$\frac{d}{dt} p_t = p_t Q$$

$$= Q p_t$$

where

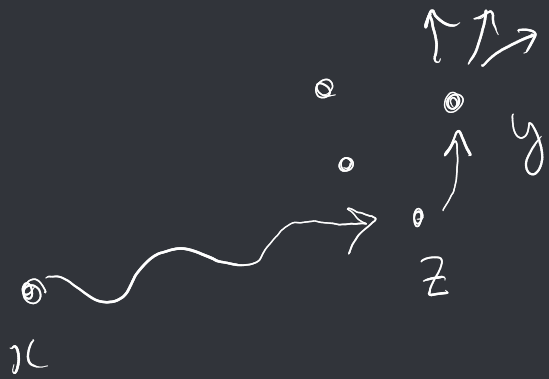
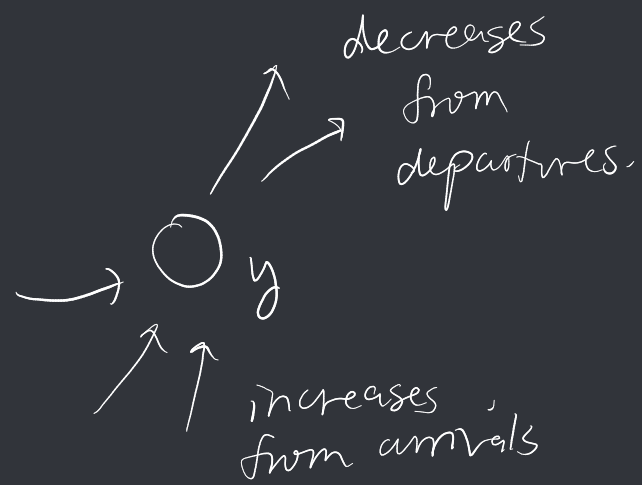
Q is transition rate matrix

$$Q_{xy} = q(x,y) \qquad y \neq x$$

$$Q_{xx} = -\lambda(x)$$

entrywise: forward equation

$$\frac{d}{dt} p_t(x,y) = \sum_{z \neq x} \overbrace{p_t(x,z)\, q(z,y)}^{\text{arriving at } y} - p_t(x,y)\,\lambda(y)$$



leaving y